

Trigonometric integrals $\int \tan^m(x) \sec^n(x) dx$

Cases:

① n is even

② m is odd, $n \geq 1$.

③ only sec ($m=0$), n is odd.

④ only tan ($n=0$)

⑤ m is even, n is odd

Canvas

$$\int \sec^4(x) dx$$

$$= \int \underbrace{\sec^2(x)}_{u^2+1} \underline{\underline{\sec^2(x) dx}}$$

$$= \int \underbrace{(\tan^2(x) + 1)}_{u^2+1} \underline{\underline{\sec^2(x) dx}}_{du}$$

$$u = \tan(x)$$

$$du = \sec^2(x) dx$$

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

$$\underline{\underline{\tan^2(x) + 1 = \sec^2(x)}}$$

$$= \int (u^2 + 1) \underline{\underline{du}}$$

$$= \frac{1}{3} u^3 + u + \underline{\underline{C}}$$

$$= \frac{1}{3} (\tan(x))^3 + \tan(x) + C.$$

$$\int \tan^3(x) \sec^6(x) dx$$

Exponent of secant is even:

$$\sec^2(x) = \tan^2(x) + 1$$

$$\int \tan^3(x) \sec^4(x) \underline{\underline{\sec^2(x)}} dx$$

$$= \int \tan^3(x) \underbrace{(\tan^2(x) + 1)^2}_{\tan^4(x) + \underline{\underline{2\tan^2(x)}} + 1} \sec^2(x) dx$$

$$\tan^4(x) + \underline{\underline{2\tan^2(x)}} + 1.$$

$$= \int \tan^3(x) (\tan^4(x) + 2\tan^2(x) + 1) \sec^2(x) dx$$

$$u = \tan(x)$$

$$du = \sec^2(x) dx$$

$$= \int u^3 (u^4 + 2u^2 + 1) du = \int (u^7 + 2u^5 + u^3) du$$

$$= \frac{1}{8} u^8 + \frac{2}{6} u^6 + \frac{1}{4} u^4 + C$$

$$= \frac{1}{8} (\tan(x))^8 + \frac{1}{3} (\tan(x))^6 + \frac{1}{4} (\tan(x))^4 + C$$

$$\int \tan^3(x) \sec^6(x) dx$$

tangent exponent is odd and there is at least one secant.

$$\frac{d}{dx}(\sec(x)) = \underline{\underline{\sec(x)\tan(x)}}$$

$$\int \tan^2(x) \sec^5(x) \left(\tan(x) \sec(x) dx \right)$$

$$\uparrow \tan^2(x) = \sec^2(x) - 1.$$

$$\int (\sec^2(x) - 1) \sec^5(x) (\tan(x) \sec(x) dx)$$

$$u = \sec(x)$$

$$du = \tan(x) \sec(x) dx$$

$$\int (u^2 - 1) u^5 du = \int (u^7 - u^5) du = \frac{1}{8} u^8 - \frac{1}{6} u^6 + C$$

$$= \frac{1}{8} \sec^8(x) - \frac{1}{6} (\sec(x))^6 + C$$

$$u = \sin x$$

$$u = x^2 + 1$$

$$\uparrow$$

$$\text{New variable} = f(\text{old variable})$$

$$\text{old variable} = f(\text{new variable})$$

$$\int \frac{x}{\sqrt{x^2+1}} dx$$

$$u = x^2 + 1$$

$$x = \tan \theta$$

$$\rightarrow \theta = \arctan(x)$$

$$\left\{ \begin{array}{l} \text{Pos. number} \cdot x^2 - \text{Pos. number} \\ a^2 \cdot x^2 - b^2 \end{array} \right. \rightarrow$$

$$\underline{ax = b \sec \theta}$$

$$\left\{ \begin{array}{l} \text{Pos. number} - (\text{Pos. number}) \cdot x^2 \\ b^2 = a^2 \cdot x^2 \end{array} \right. \rightarrow$$

$$\underline{ax = b \sin \theta}$$

$$\left\{ \begin{array}{l} \text{Pos. number} \cdot x^2 + \text{Pos. number} \\ a^2 \cdot x^2 + b^2 \end{array} \right. \rightarrow$$

$$\underline{ax = b \tan \theta}$$

$$\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$$

$$\boxed{x = 2 \tan \theta}$$

$$\underbrace{(ax)^2 + b^2}_{a^2 x^2} \rightarrow ax = \frac{b \tan \theta}{\sqrt{5}}$$

$$x = f(\theta)$$

$$\frac{dx}{d\theta} = f'(\theta) \Rightarrow \boxed{dx = f'(\theta) d\theta}$$

$$x^2 = 4 \tan^2 \theta$$

$$x^2 + 4 = 4 \tan^2 \theta + 4 = 4 (\tan^2 \theta + 1) = 4 \sec^2 \theta$$

$$\sqrt{x^2 + 4} = 2 \sec \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

$$\int \frac{1}{(4 \tan^2 \theta)(2 \sec \theta)} d\theta$$

$$= \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$\sec \theta \cdot \frac{1}{\tan^2 \theta} = \frac{1}{\cos \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{\cos \theta}{\sin^2 \theta}$$

$$= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta = \frac{1}{4} \int u^{-2} du = \frac{1}{4} \frac{u^{-1}}{-1} + C$$

$$u = \sin \theta$$

$$du = \cos \theta d\theta$$

$$= -\frac{1}{4} \cdot \frac{1}{u} + C$$

$$\tan \theta = \frac{x}{2}$$

$$\theta = \arctan\left(\frac{x}{2}\right)$$

$$\uparrow \text{I, IV quadrant}$$

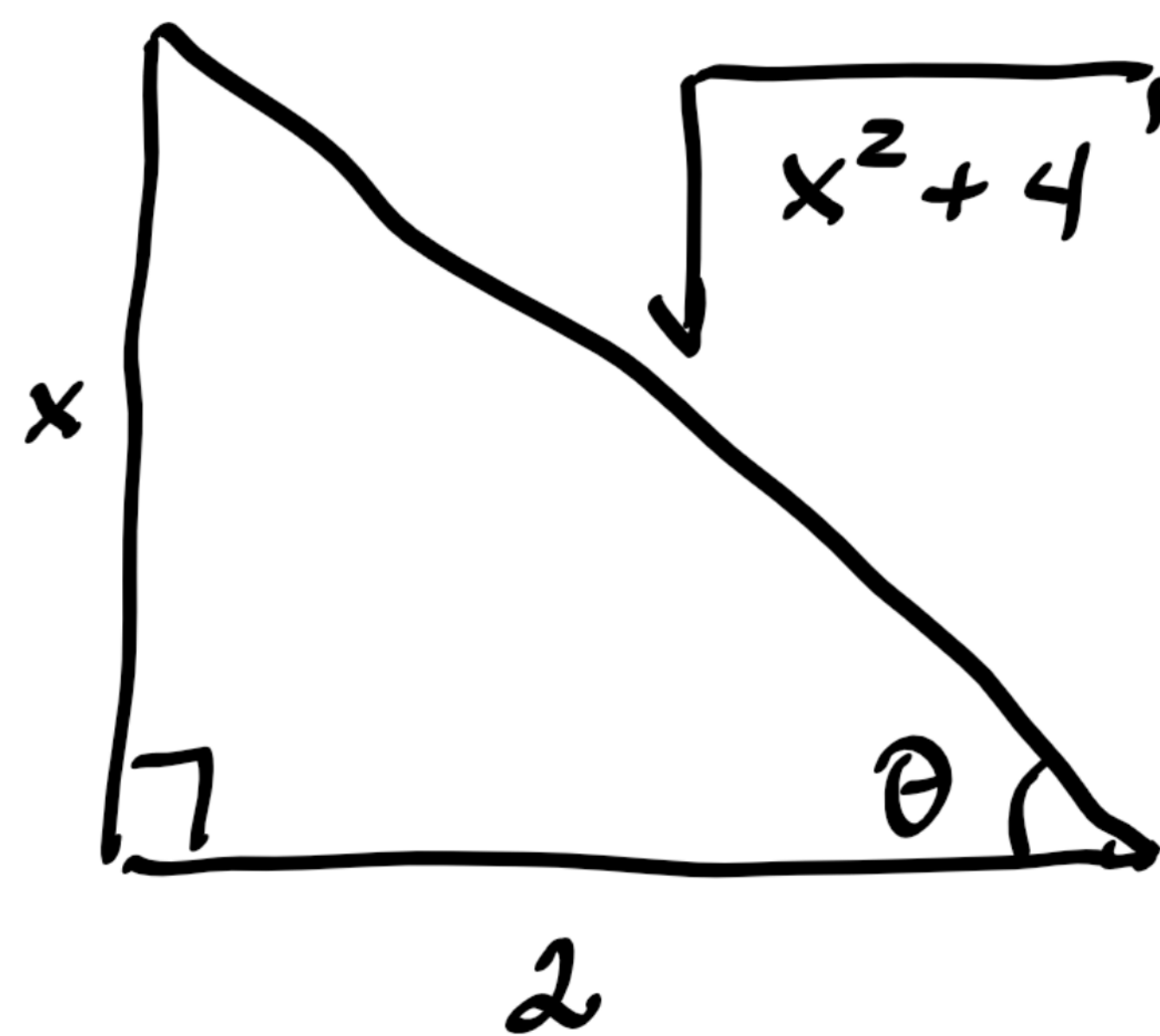
$$\sec(\theta) > 0$$

$$= -\frac{1}{4} \cdot \frac{1}{\sin \theta} + C$$

$$= -\frac{1}{4} \frac{\sqrt{x^2+4}}{x} + C$$

$$\underline{x = 2 \tan \theta}$$

$$\tan \theta = \frac{x}{2}$$



$$\sin \theta = \frac{x}{\sqrt{x^2+4}}$$

$$\int_{-1/2}^{\sqrt{2}/2} \frac{x^3}{\sqrt{1-x^2}} dx$$

$$-\frac{1}{2} = \sin \theta \Rightarrow \theta = -\pi/6$$

$$\frac{\sqrt{2}}{2} = \sin \theta \Rightarrow \theta = \pi/4$$

$$x = \sin \theta$$

$$x^3 = \sin^3 \theta$$

$$1 - x^2 = 1 - \sin^2 \theta = \cos^2 \theta$$

$$\sqrt{1-x^2} = |\cos \theta| = \cos \theta$$

$$dx = \cos \theta d\theta$$

$$x = \sin \theta$$

$$\theta = \arcsin(x)$$

1st, 2nd quadrant $\Rightarrow \cos(\theta) \geq 0$

$$\int \frac{\sin^3 \theta}{\cos \theta} \cdot \cos \theta d\theta$$

$$= \int \sin^3 \theta d\theta = \int \sin^2 \theta (\sin \theta d\theta) = \int (1 - \cos^2 \theta) (\sin \theta d\theta)$$

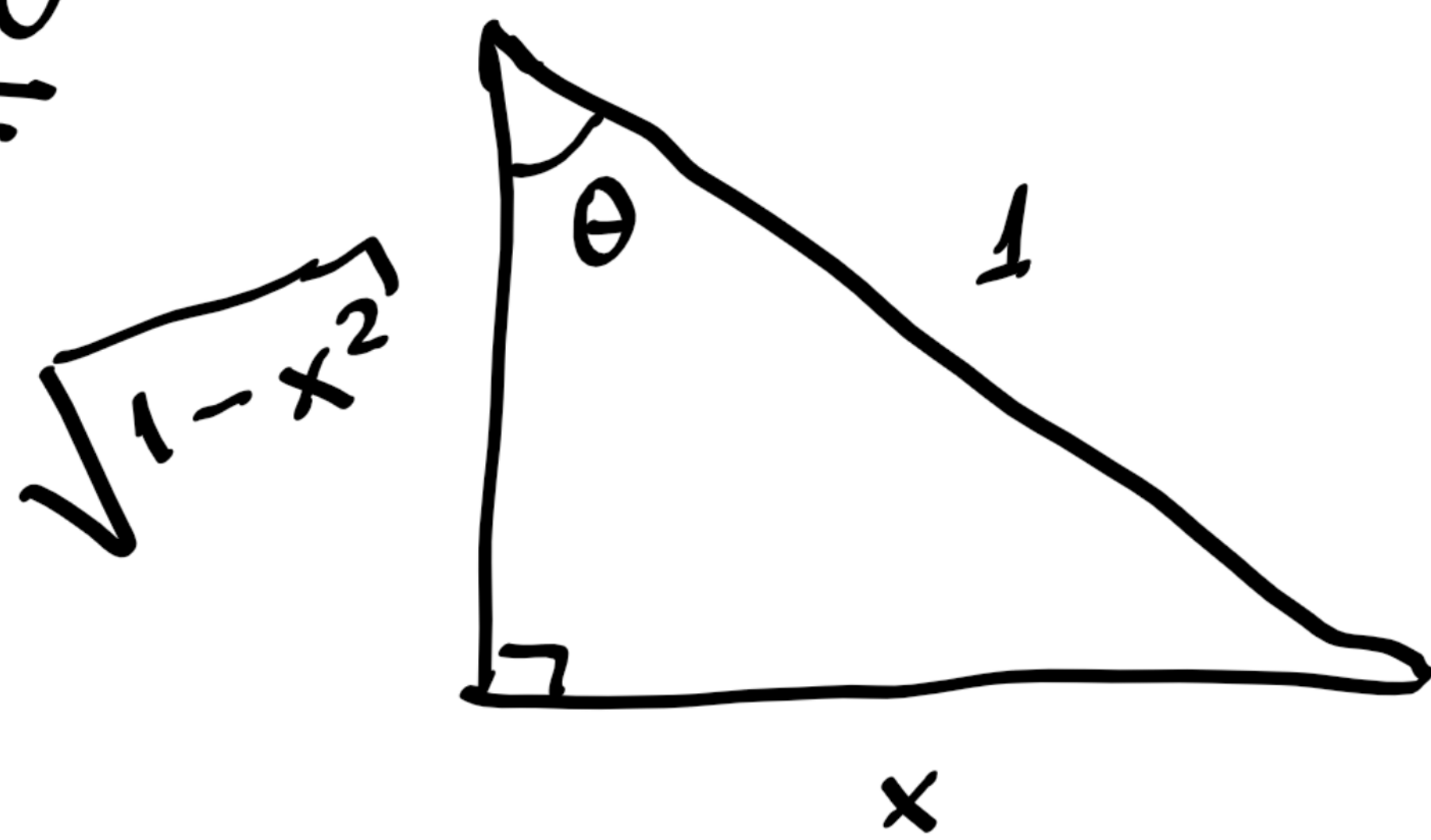
$$u = \cos \theta$$

$$du = -\sin \theta d\theta$$

$$-\int (1 - u^2) du = \int (u^2 - 1) du$$

$$= \frac{u^3}{3} - u + C = \frac{1}{3}(\cos \theta)^3 - \cos \theta + C$$

$$\underline{\underline{x = \sin \theta}}$$



$$\cos \theta = \sqrt{1-x^2}$$

$$= \frac{1}{3} \left(\sqrt{1-x^2} \right)^3 - \sqrt{1-x^2} + C$$

$$\frac{1}{3} \left(\sqrt{1 - \left(\frac{\sqrt{2}}{2} \right)^2} \right)^3 - \sqrt{1 - \left(\frac{\sqrt{2}}{2} \right)^2} - \left(\frac{1}{3} \left(\sqrt{1 - \left(-\frac{1}{2} \right)^2} \right)^3 - \sqrt{1 - \left(-\frac{1}{2} \right)^2} \right).$$

$$\int \frac{x^2}{\sqrt{x^2-2}} dx$$

$$\underline{\underline{x^2 = 2 > 0}}$$

$$\textcircled{x > \sqrt{2}}$$

$$\text{or } \underline{\underline{x < -\sqrt{2}}}$$

Easy

$$\textcircled{x = \sqrt{2} \sec \theta}$$

$$x^2 = 2 \sec^2 \theta$$

$$x^2 - 2 = 2 \sec^2 \theta - 2 = 2 (\sec^2 \theta - 1) = 2 \tan^2 \theta$$

$$\sqrt{x^2 - 2} = \sqrt{2} |\tan \theta| \quad \begin{cases} \sqrt{2} \tan \theta \rightarrow \text{positive } x \\ -\sqrt{2} \tan \theta \rightarrow \text{negative } x \end{cases}$$

$$\underline{dx = \sqrt{2} \tan \theta \cdot \sec \theta d\theta}$$

$$\int \frac{2 \sec^2 \theta}{\cancel{\sqrt{2} \tan \theta}} \cdot \cancel{\sqrt{2} \tan \theta} \sec \theta d\theta = 2 \int \sec^3 \theta d\theta$$

$$ax = b \tan \theta \rightarrow \frac{a dx}{a} = \frac{b \sec^2 \theta}{a} d\theta \rightarrow dx = \frac{b}{a} \sec^2 \theta d\theta$$

$$ax = b \sec \theta \rightarrow \frac{a dx}{a} = \frac{b \sec \theta \tan \theta}{a} d\theta \rightarrow dx = \frac{b}{a} \sec \theta \tan \theta d\theta$$

$$ax = b \sin \theta \rightarrow \frac{a dx}{a} = \frac{b \cos \theta}{a} d\theta \rightarrow dx = \frac{b}{a} \cos \theta d\theta$$

$$2 \int \sec^3 \theta d\theta = 2 \int \sec \theta \sec^2 \theta d\theta$$

$$u = \sec \theta \quad v = \tan \theta$$

$$du = \sec \theta \tan \theta d\theta \quad dv = \sec^2 \theta d\theta$$

$$= 2 \left[\sec \theta \tan \theta - \int \sec \theta \tan^2 \theta d\theta \right]$$